P3: GreenStep

– Personal Carbon Footprint & Eco Lifestyle Tracker –

"Small daily steps. Measurable climate impact."

PR1: Group Project Proposal

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I. Problem Framing: Statement & Objectives

1. Background & Context

Climate change represents a critical and complex global crisis, with its underlying causes intricately linked to individual behavioral patterns. A substantial proportion of global greenhouse gas emissions can be traced back to the cumulative choices made by individuals in their daily lives, particularly in areas such as transportation, dietary preferences, and energy consumption within households. Everyday decisions, such as opting for personal gasoline-powered vehicles instead of public transportation, choosing carbon-intensive red meat over plant-based alternatives, or utilizing energy-intensive appliances contributing significantly to our overall environmental impact.

To meet international climate targets, it is imperative to facilitate a widespread shift towards more sustainable lifestyles. However, the challenge remains that establishing enduring sustainable habits can be problematic. One of the key obstacles is that individuals often lack immediate, quantifiable feedback regarding the carbon implications of their everyday actions. This absence of clear metrics hampers the ability to connect individual behavior with broader environmental outcomes, thereby complicating efforts to foster lasting change toward sustainability.

2. The Core Problem

Despite a growing public interest in eco-friendly living, existing climate awareness and carbon-tracking mobile applications fail to retain long-term user engagement due to a failure to strike a balance between usability and empirical accuracy:

- **Oversimplification:** Many applications function merely as static text repositories for generic, non-quantifiable "green tips", such as "turn off the lights". They lack personalized data tracking, automated calculations, and interactive feedback loops, which ultimately fails to motivate habit-building.

- **Overcomplication:** Conversely, scientific carbon footprint calculators overwhelm everyday users by demanding exhaustive, granular Life-Cycle Assessments (LCAs). Forcing users to manually input exact monthly utility bills, hyper-precise travel distances, and complex grocery receipts introduces massive cognitive load, resulting in user frustration, rapid burnout, and high app abandonment rates.

Currently, a structural gap exists for a cross-platform solution that provides empirical, scientifically grounded tracking through a low-friction, highly engaging interface.

3. Proposed Solution

GreenStep is an innovative cross-platform tool for tracking personal carbon footprints and promoting eco-friendly lifestyles. The application is constructed with a lightweight **Vue 3** frontend, backed by a **PHP** Slim 4 **RESTful** API, and utilizes a **MySQL** database for data management. It converts the process of environmental accountability into a seamless daily practice, allowing users to log core activities such as transportation (in kilometers), dietary choices, electricity consumption (in kWh), and recycling with minimal friction through single-tap inputs.

In order to foster long-term engagement, GreenStep leverages behavioral psychology principles, incorporating features like gamified daily streaks and digital badges, alongside a randomized library of daily eco-tips. This enhances user motivation and retention. The platform also adopts a unique multi-role architecture, distinguishing between **Users**, **Community Leaders**, and **Administrators**. This structure fosters community involvement, enabling leaders to create challenges centered on collective carbon reduction goals, thereby transforming individual sustainability efforts into collaborative social initiatives.

To ensure high data integrity, all backend carbon calculations are meticulously decoupled from the application's source code. These calculations are dynamically aligned with verified institutional baselines, such as those provided by **DEFRA** and the **IPCC**, with oversight facilitated through a secure administrative control panel. This architecture not only enhances the credibility of the data but also promotes transparency and trust in the carbon tracking process.

4. Project Objectives

GreenStep Development Objectives is to ensure a secure, efficient, and production-ready platform, the development of GreenStep adheres to the following clearly defined technical and functional goals:

- **Objective 1: Multi-Role Architecture & Secure Access Control**

Implement a role-based access control system that delineates permissions for Environmentally Curious Individuals, Community Leaders, and Platform Administrators. All endpoints will be secured using stateless **JSON Web Tokens (JWT)**, while user credentials will be safeguarded through industry-standard password hashing techniques, thereby ensuring robust data segregation.

- **Objective 2: High-Performance Server-Side Calculation Engine**

Construct a modular RESTful API leveraging the **PHP Slim 4** micro-framework to serve as the authoritative computation engine. This engine will be responsible for dynamically processing user logs against database-driven emission factors, utilizing **PHP Data Objects (PDO)** and parameterized prepared statements to effectively mitigate SQL injection risk.

- **Objective 3: Low-Friction, Cross-Platform Interface**

Design a responsive, mobile-first user interface as a Vue 3 Single Page Application (SPA) with Pinia for centralized state management. The web application will be encapsulated with **Capacitor** to create a native Android application package (APK), ensuring that users can log activities or access their impact dashboards within a 60-second window.

- **Objective 4: Interactive Data Visualization**

Incorporate **Chart.js** into the frontend dashboard to convert raw backend JSON payloads into dynamic visualizations of weekly and monthly trends, providing users with transparent historical insights into their carbon reduction journeys.

- **Objective 5: Gamification & Social Retention Loops**

Implement an automated engagement engine that tracks user activity streaks, unlocking achievement badges based on structured JSON criteria. The system will also distribute context-aware eco-tips and aggregate collective data metrics for community challenges to enhance user engagement and retention.

II. Target Users & Scope

1. Target User Personas

GreenStep is designed around three distinct user personas, each interacting with the system through a dedicated role and a clearly bounded permission scope;

1.1. Environmentally Curious Individuals

This is the platform's primary end-user; an everyday person who wishes to understand and reduce their personal carbon footprint without possessing specialist environmental expertise. This user requires a low-friction daily logging experience that converts ordinary activities (commuting, meals, household electricity, recycling) into a comprehensible carbon estimate. Within the system, this persona authenticates through a standard end-user account, submits daily activity logs, views a personalized dashboard with historical charts, earns streaks and badges, and opts into community challenges.

1.2. The Community / Group Leader is an organizer

For example, the head of a residential college block, a student society, or a workplace green team who coordinates collective sustainability efforts. This persona's needs are motivational and administrative at the group level: creating and configuring community challenges, defining collective reduction targets, and monitoring aggregated group progress. Within the system, they retain all end-user capabilities while additionally provisioning challenges and tracking member participation against shared goals.

1.3. The Platform Administrator

This is the trusted operator responsible for the integrity and credibility of the platform's data. This persona curates the daily tip library, maintains the per-activity emission factors that underpin every carbon calculation, and verifies challenge results. Within the system, the administrator holds an elevated, role-gated account with exclusive access to the management endpoints governing tips, emission factors, and challenge validation; a deliberate separation ensuring that end-users can never alter the scientific basis of the calculation engine.

2. Product Scope

The product scope for GreenStep is strategically divided into two tiers to ensure a structured, iterative development cycle. This approach balances the immediate academic requirements of the project with a vision for future user engagement and enhanced platform capabilities.

2.1. Must-Have Features

This set defines the minimum viable product required to satisfy the course's core learning outcomes (CLO) and deliver a coherent end-to-end experience;

- **Secure Authentication:** User registration and login utilizing hashed passwords and JWT-protected endpoints across all three roles.
- **Core Activity Logging:** A dedicated interface to capture daily transport (mode/distance in km), meals (vegetarian/mixed/red meat), electricity consumption (kWh), and recycling counts.
- **Server-Side Calculations:** An authoritative backend calculation engine that processes user inputs against configurable, per-activity emission factors.
- **Interactive Dashboard:** A responsive Vue 3 frontend surfacing user results through weekly and monthly trend visualizations via Chart.js.
- **Gamification Engine:** A streak-and-badge system rewarding consecutive low-footprint days, alongside a randomized daily eco-tip library and group challenges with collective targets.
- **Role-Based Administration:** Gated administrative access for curating tips and modifying core emission factors securely.

- **Cross-Platform Delivery:** A mobile-first design approach wrapped as a deployable native Android build using Capacitor (camera access reserved as optional for proof-of-action).

2.2. Should-Have Features

This set comprises high-value enhancements that deepen engagement and social motivation, but are not strictly required for the initial functional product:

- **Social Connections:** Friend connectivity paired with a community leaderboard to introduce friendly competition.
- **Photo Verification:** Optional photo uploads for verified eco-actions, such as using a reusable cup, that award bonus points and reinforce accountability.
- **Engagement Notifications:** Daily reminders; delivered in-app or through Capacitor's local notification API, to sustain consistent user logging habits.

III. Technology Stack & Architecture

1. Frontend Tier

GreenStep's frontend tier will be developed as Vue 3 Single Page Application (SPA) using a component-based architecture to ensure maintainability, scalability, and responsive cross-platform usability.

The frontend application will utilize Vue Router to manage navigation between pages. Pinia will be used for centralized state management, including authentication tokens, user session data, dashboard statistics, activity logs, and challenge participation information.

The user interface will follow a mobile-first responsive design approach, ensure compatibility across smartphones, tablets, and desktop devices. For common interface elements, such as navigation bars, activity cards, forms, charts, eco tips, badges, and challenge items, we will implement Reusable Vue Single File Component (SFCs).

For asynchronous communication with the PHP Slim 4 backend API, Axios or Fetch API will be used. JSON responses from the backend will be dynamically rendered in the dashboard, activity logs, and challenge pages without requiring full-page reloads.

To visualize weekly and monthly carbon footprint statistics, integrated chart.js into the Vue application, design it in an interactive and user-friendly format.

Accessibility considerations such as semantic HTML structure, readable typography, and keyboard navigability will also be incorporated to improve usability for all users.

2. Backend Tier

The backend tier of GreenStep is implemented as a lightweight, modular RESTful API built on the PHP Slim 4 micro-framework. The architecture enforces a strict separation of concerns, decoupling route definitions, business logic (controllers), middleware, and output generation. This tier serves as the central data and computation engine for all cross-platform clients.

The API operates entirely through JSON request and response shaping, consuming structured application/json bodies and emitting consistently structured JSON payloads for predictable consumption by the Vue 3 frontend. A vital responsibility of this tier is rigorous error handling: a centralised Slim error handler intercepts every logic exception, routing failure, and server fault, transforming it into a unified JSON error envelope paired with a semantically precise HTTP status code, such as 200 OK for success, 201 Created for resource instantiation, 400 Bad Request for malformed input, 401 Unauthorized for a missing or invalid token, 403 Forbidden for an authenticated user lacking the required role, 404 Not Found for absent resources, and 500 Internal Server Error for unhandled exceptions.

Access control is enforced by a JWT authentication middleware applied to every protected route; each token carries the user's identity and role claim, enabling role-based gating that separates end-users, community leaders, and administrators. All persistence is handled through PHP Data Objects (PDO) using prepared statements exclusively, providing a structural defence against SQL injection at the data-access boundary as mentioned in the Database Layer *(Section III.3)*.

As the definitive contract between backend logic and the frontend UI, all endpoints follow strict, self-describing RESTful naming conventions. The table below presents the initial API contract, demonstrating full CRUD coverage across the two primary entities; Activity Logs and Challenges, alongside the authentication and administrative routes:

Endpoint Route	HTTP Method	Expected JSON Payload (Input)	Role Context	Success Code
/api/auth/register	POST	{ name, email, password }	Public	201
/api/auth/login	POST	{ email, password }	Public	200
/api/logs	GET	None	Authenticated User	200
/api/logs	POST	{ activity_type_id, amount, logged_on }	Authenticated User	201
/api/logs/{id}	PUT	{ activity_type_id, amount, logged_on }	Authenticated User	200
/api/logs/{id}	DELETE	None	Authenticated User	200
/api/dashboard	GET	None	Authenticated User	200
/api/challenges	GET	None	Authenticated User	200
/api/challenges	POST	{ name, description, start_date, end_date, target_co2_reduction }	Community Leader	201
/api/challenges/{id}	PUT	{ name, description, target_co2_reduction }	Community Leader	200
/api/challenges/{id}	DELETE	None	Community Leader	200
/api/challenges/{id}/join	POST	None	Authenticated User	201
/api/tips/daily	GET	None	Authenticated User	200
/api/admin/tips	POST	{ title, body, category }	Administrator	201
/api/admin/factors/{id}	PUT	{ kg_co2_per_unit }	Administrator	200

3. Database Layer

3.1. Core Database Management System (MySQL)

The application will utilize MySQL as its persistent relational database management system MySQL was selected for its robustness, ACID compliance, and efficiency in handling structured relational data. It will reliably manage the complex relationships between the system's core entities such as mapping individual users to their daily activity logs, tracking community challenge membership, and storing the dynamic carbon emission multipliers (Activity Types).

3.2. Data Access Layer (PHP Data Objects - PDO)

Communication between the PHP Slim 4 backend and the MySQL database will be handled exclusively through PHP Data Objects (PDO). PDO provides a lightweight, consistent, and object-oriented interface for database interactions. By utilizing PDO the application benefits from centralized database connection management, streamlined error handling (via PDO exceptions), and a clean separation of concerns between the business logic and data storage layers.

3.3. Security Protocols & Prepared Statements

Security and data integrity are paramount especially when handling user credentials and tracking data. To provide robust defense mechanisms against SQL Injection (SQLi) attacks the application strictly prohibits dynamic string interpolation in SQL queries instead, all database interactions executed via the PDO layer will exclusively utilize **prepared statements and parameterized queries**. This security measure ensures that any data sent by the client is strictly evaluated as data payloads rather than executable SQL commands.

3.4. Data Integrity & Constraints

Furthermore, the database layer enforces referential integrity at the schema level. Foreign key constraints will be rigorously applied (e.g., automatically cascading deletes from the **users** table to their respective **activity_logs** or **challenge_memberships**) to prevent orphaned records and ensure the database remains in a consistent state throughout the application's lifecycle.

4. Cross-Platform Wrapper & Deployment

4.1. Cross-Platform Mobile Wrapper (Capacitor)

To fulfill the mobile adaptability requirement of the course, we will utilize **Capacitor** to wrap our responsive Vue 3 web application into a deployable native Android application (APK). Capacitor serves as a seamless bridge between our web codebase and the native mobile environment. This approach allows us to maintain a single unified codebase for both web and mobile users while granting the GreenStep app potential access to native device features (such as the camera, should users need to upload photo proof of their daily eco-friendly activities).

4.2. Cloud Hosting & Public Deployment

The GreenStep system will be deployed to a live cloud environment to ensure public accessibility, allowing for real-world testing and end-to-end demonstration during the final PR4 presentation. The deployment architecture is divided as follows:

- **Frontend Hosting:** The compiled Vue 3 Single Page Application (SPA) will be hosted on a modern static site host (such as Vercel, Netlify, or GitHub Pages) to ensure fast, reliable delivery of the user interface.
- **Backend API & Database Hosting:** The PHP Slim 4 RESTful API and the MySQL database will be deployed on a free-tier cloud hosting provider (such as InfinityFree, Render, or a similar shared hosting environment that supports PHP and PDO).
- **Version Control Integration:** We will use GitHub to manage our source code, establishing a basic deployment pipeline where pushes to the repository can seamlessly update our live frontend environment.

IV. UI/UX Mock-Ups

1. Interface Design Systems

The GreenStep interface design follows a clean and eco-friendly visual identity using green, white, and natural color tones to promote sustainability awareness.

The application adopts a mobile-first responsive layout, using consistent typography, rounded cards, reusable buttons, navigation bars, and dashboard components to maintain a unified user experience.

2. Screen-by-Screen Wireframes

Hereby are the app mock up interfaces in the mobile app view:

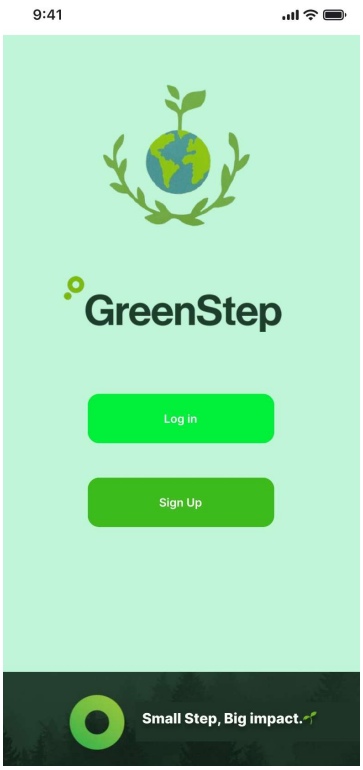


Figure 1: Get Started Interface

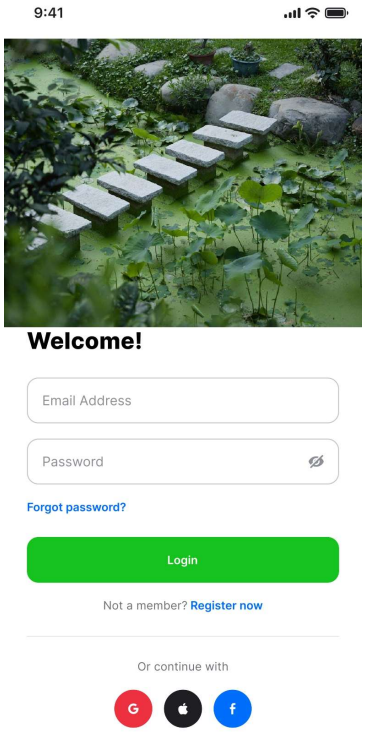


Figure 2: Login Interface

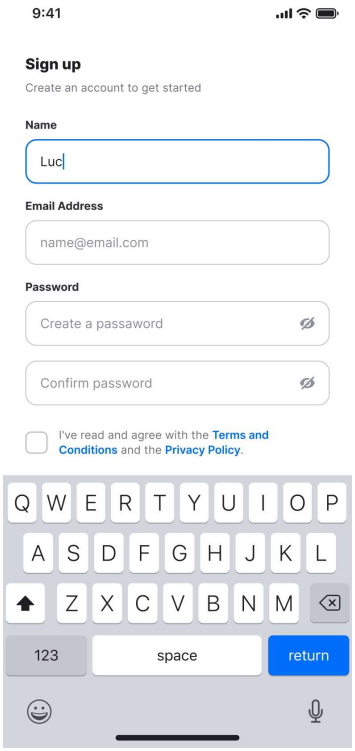


Figure 3: Sign up Interface

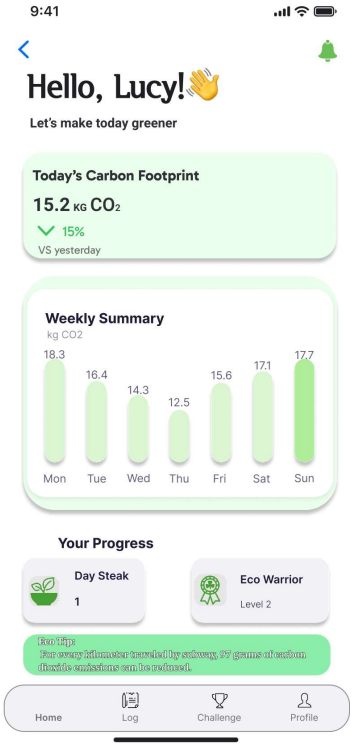


Figure 4: User Home Page Interface

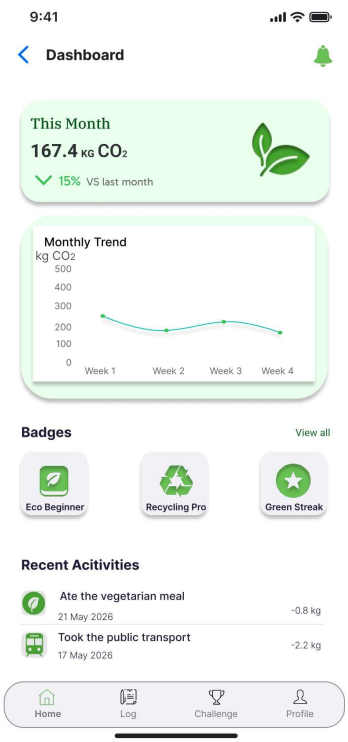


Figure 5: User Dashboard Interface

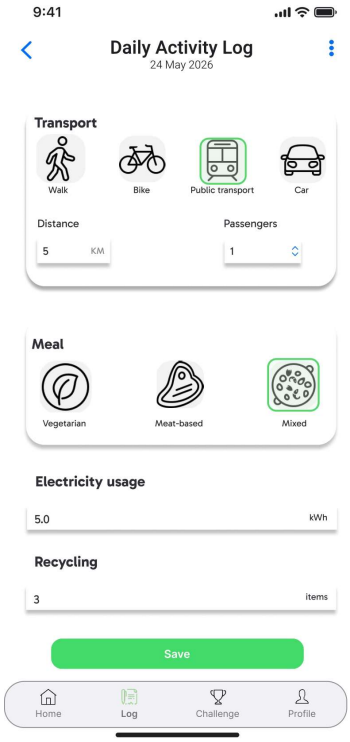


Figure 6: Daily Activity Log Interface

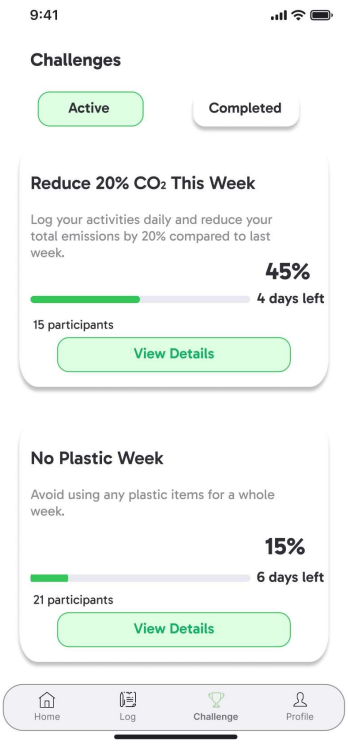


Figure 7: Challenges Interface

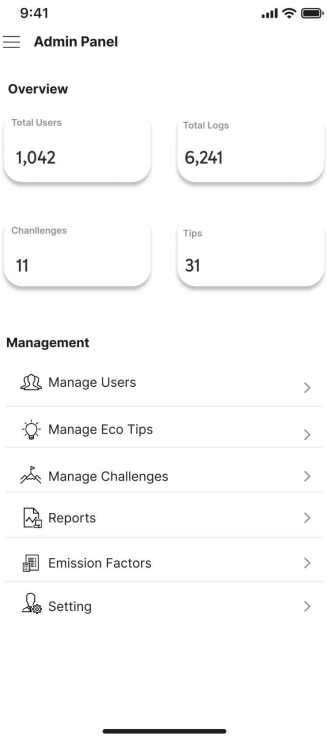


Figure 8: Admin Panel Interface

The following is the administrative dashboard interface in the desktop view:

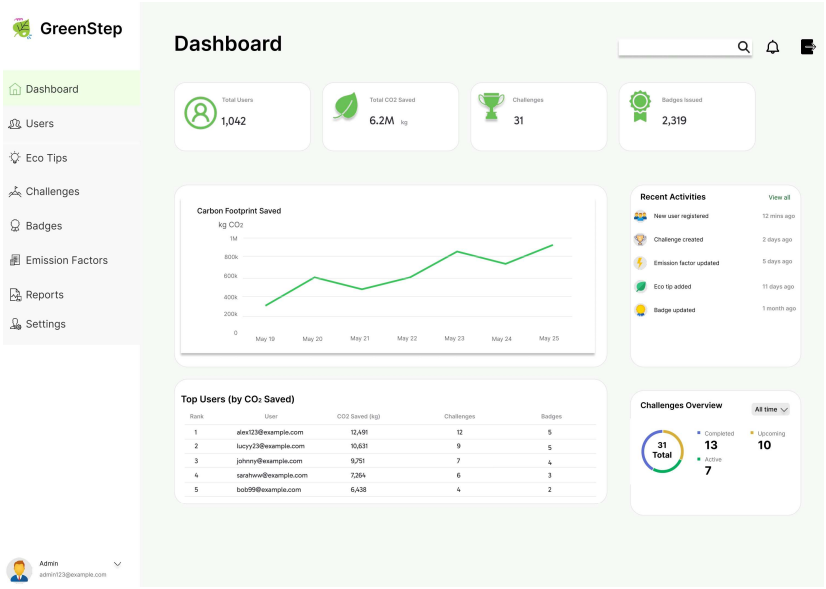


Figure 9: Administrative Dashboard (Desktop Interface)

V. Entity-Relationship Diagram (ERD) & Suggested Data Model

1. Conceptual Database Design

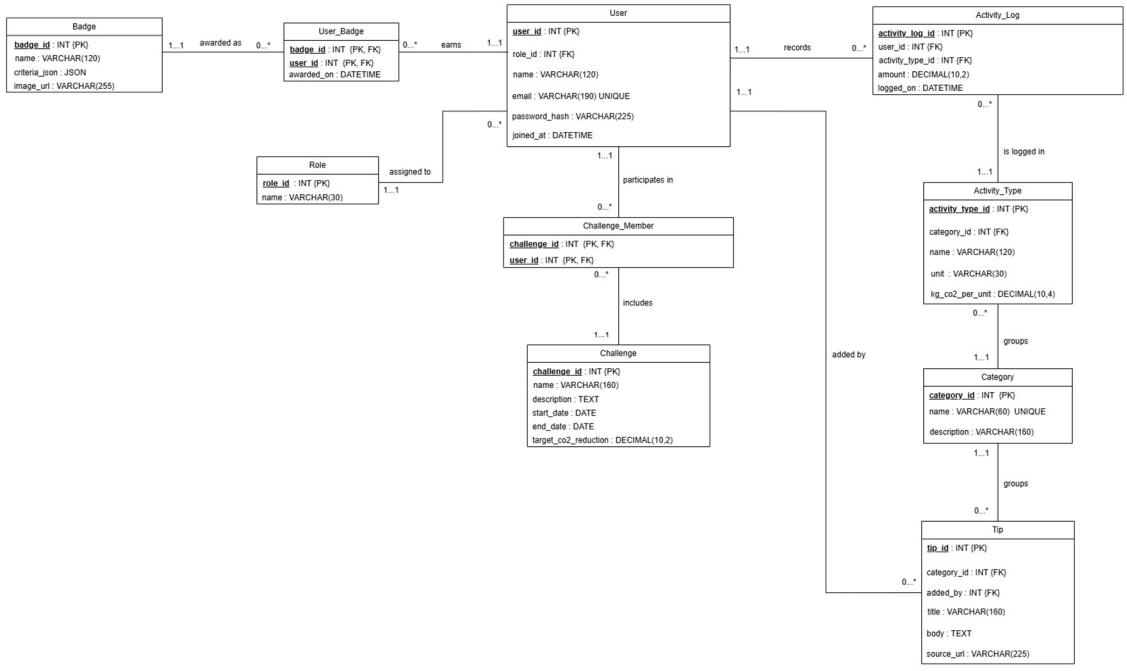


Figure 10: ERD for the proposed solution

2. Data Dictionary / Schema Specifications

2.1. Description of entities

Entity	PK	FK / Links	Description	Occurrence
User	user_id : INT	role_id : INT / User	Stores user credentials, email, and role-based access policies.	Each user can record zero or more activity logs and participate in zero or more challenges.
Activity_Type	activity_type_id : INT	category_id : INT / Category	Stores tracking metrics and configurable per-activity emission factors.	Each activity type can be referenced by zero or more activity logs.
Activity_Log	activity_log_id : INT	user_id : INT / User activity_type_id : INT / Activity_Type	Stores individual daily footprint logs mapped to users and specific activities.	Each activity log belongs to exactly one user and exactly one activity type.
Tip	tip_id : INT	category_id : INT / Category added_by : INT/ User	Stores the library of randomised daily eco-tips curated by administrators.	Each tip is a standalone record accessed globally by the system.

Challenge	challenge_id : INT	(no FK)	Stores group goals, durations, and collective carbon reduction targets.	Each challenge can include zero or more challenge members.
Challenge_Member	challenge_id : INT user_id : INT	challenge_id : INT / Challenge user_id : INT / User	Stores the participation junction details between users and challenges.	Each challenge member record corresponds to exactly one challenge and exactly one user.
Badge	badge_id : INT	(no FK)	Stores gamification awards and criteria JSON needed for streaks.	Each badge is a standalone record accessed for gamification logic.
Roles	role_id : INT	(no FK)	Stores user role designations (e.g., admin, user).	Each role can be assigned to zero or more users.
User_Badges	badge_id : INT user_id : INT	badge_id : INT / Badge user_id : INT / User	Stores the junction details tracking which users have earned which badges.	Each user badge record corresponds to exactly one user and exactly one badge

2.2. Description of relationships

Entity 1	Multiplicity	Relationship	Entity 2	Multiplicity
User	0..*	Has	Roles	1..1
User	1..1	Recods	Activity_Log	0..*
User	1..1	Participates	Challenge_Member	0..*
User	1..1	Earns	User_Badges	0..*
Activity_Type	1..1	Categories	Tip	0..*
Activity_Type	1..1	Logged in	Activity_Logs	0..*
Activity_Log	0..*	Is recorded by	Users	1..1
Activity_Log	0..*	Refers to	Activity_Types	1..1
Tip	0.. *	Categories	Activity_Type	1..1
Challenge	1..1	Includes	Challenge_Members	0..*
Challenge_Member	0..*	Belongs to	Challenge	1..1
Challenge_Member	0..*	Belongs to	User	1..1
Badge	1..1	Awarded as	User_Badges	0..*
Roles	1..1	Assigned to	User	0..*
User_Badges	0..*	Awarded to	User	1..1
User_Badges	0..*	Refers to	Badge	1..1

3. Security & Data Integrity Policies

To ensure platform credibility and safeguard sensitive user data, strict backend and database policies will be enforced:

- **SQL Injection (SQLi) Defense:** All database queries executed by the PHP backend will strictly utilize PHP Data Objects (PDO) with **prepared statements and parameterized queries**. Under no circumstances will dynamic user input be directly concatenated into SQL strings, preventing malicious execution of database commands.

- **Cross-Site Scripting (XSS) Prevention:** Output escaping will be meticulously applied. While the Vue 3 frontend automatically escapes HTML by default to prevent client-side script injection, the PHP API will also sanitize user inputs (using functions like htmlspecialchars()) before data has persisted or echoed back in JSON responses, ensuring comprehensive defense-in-depth.
- **Strict JWT Role Validation:** Authentication and authorization will be handled statelessly via JSON Web Tokens (JWT). The backend will enforce strict middleware checks verifying the signature and expiration of the JWT on every protected route. Furthermore, role-based policies embedded within the JWT payload will be validated to ensure standard users cannot access admin endpoints (e.g., preventing regular users from modifying the kg_co2_per_unit emission factors).
- **Relational Data Integrity:** The MySQL schema will employ strict foreign key constraints (e.g., ON DELETE CASCADE). If a user account or a community challenge is deleted, all dependent records (such as activity logs or challenge memberships) are automatically removed, preventing orphaned records and database anomalies.

VI. Entrepreneurial Angle

1. Revenue Models

To transform GreenStep from a standard tracker into a self-sustaining and commercially viable cross-platform product, we propose a two-pronged monetization strategy:

1.1. Corporate B2B SaaS Tier

Organizations and universities are increasingly seeking tangible ways to meet their environmental, social, and governance (ESG) goals and boost employee engagement. GreenStep will offer a paid business-to-business (B2B) subscription model allowing companies to establish private, closed-loop sustainability challenges for their workforce. Corporate administrators will receive a dedicated web dashboard to configure team targets, view aggregated (and anonymized) carbon reduction analytics for sustainability reporting, and incentivize employee participation.

1.2. Affiliate Brand Partnerships for Reward Fulfillment

To keep the consumer version free and engaging, user gamification will be tied to real-world rewards. By maintaining consecutive low-footprint streaks or successfully completing community challenges, users earn points. These points can be redeemed for exclusive discount codes, vouchers, or eco-friendly merchandise provided by our vetted sustainable brand partners. This generates revenue via corporate partnership placement fees and affiliate marketing commissions, creating a win-win ecosystem where green lifestyle habits are actively rewarded.

2. Trust & Data Credibility

User adoption and long-term retention rely heavily on the perceived accuracy of our environmental tracking. Vague estimations degrade user trust. To ensure our server-side carbon calculation logic is indisputable and reliable, the team will implement the following strategies:

2.1. Standardized Government & Institutional Baselines

Our database's ActivityType emission factors will be strictly sourced from globally recognized and peer-reviewed environmental frameworks. For initial deployment, parameters will align with standard carbon data values from public frameworks, such as the UK Department for Environment, Food & Rural Affairs (DEFRA) or the Intergovernmental Panel on Climate Change (IPCC).

2.2. Decoupled Configuration & Verification API Design

Instead of hardcoding calculation multipliers into our PHP Slim 4 source files, emission factors will be dynamically maintained in a relational database table controlled entirely via an authenticated Admin module. As an innovation milestone, the backend will be architected to pull or reference live open-data registries via RESTful endpoints, ensuring that values automatically adjust as regional energy grids shift toward renewable resources.

2.3. Transparent Methodology and Proof Validation

Trust is reinforced through transparency. Every calculation summary on the Vue 3 dashboard will include a breakdown of the specific emission formula used. Furthermore, to verify community challenge entries, the mobile build will leverage native device features using Capacitor, enabling an optional camera proof upload mechanism so users can structurally verify their high-impact eco-actions.

VII.7-week Milestone Plan

1. Weekly Development Timeline

The following timeline maps out our 7-week active development sprint for the GreenStep application, covering the full syllabus pipeline from cross-platform web architecture (Chapter 1) to mobile deployment (Chapter 13).

Phase no.	Phase	Task / Syllabus Pipeline	Lead Role	Team Member	Start Week	End Week	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
1	Planning, Proposal & Environment Setup	Draft PR1 Project Proposal	All Members	Group 2	Week 1	Week 1							
		Design MySQL Database Schema & ER Diagram	Database & Security Lead	Member 3 - Rawan	Week 1	Week 1							
		Initialize Git Repo & DevOps Environment	DevOps, Mobile & Integration Lead	Member 4 - Zineb	Week 1	Week 1							
		Design Mobile-First UI Mockups	Frontend Lead	Member 1 - ZengLiting	Week 1	Week 2							
		Setup PHP Slim 4 Backend & API Contract	Backend & API Lead	Member 2 - Mohammed	Week 2	Week 2							
2	Static UI & Interim Build Preparation	Build Static HTML/CSS Interfaces	Frontend Lead	Member 1 - ZengLiting	Week 2	Week 3							
		Implement PDO Data-Access Layer	Database & Security Lead	Member 3 - Rawan	Week 2	Week 3							
		Create Mock JSON API & Fetch Data (PR2 Goal)	Backend & API Lead	Member 2 - Mohammed	Week 3	Week 3							
		Prepare & Submit PR2 Interim Build	All Members	Group 2	Week 3	Week 3							
3	Vue Scaffold & Core API Development	Scaffold Vue 3 SPA & Vue Router	Frontend Lead	Member 1 - ZengLiting	Week 3	Week 4							
		Refactor Static UI into Vue SFCs	Frontend Lead	Member 1 - ZengLiting	Week 4	Week 5							
		Develop RESTful CRUD Endpoints	Backend & API Lead	Member 2 - Mohammed	Week 4	Week 5							
		Apply Database Security Measures	Database & Security Lead	Member 3 - Rawan	Week 4	Week 5							
4	Advanced Logic & System Integration	Implement Server-Side Carbon Calculation Logic	Backend & API Lead	Member 2 - Mohammed	Week 5	Week 6							
		Integrate JWT Auth & Role-Based Access	Database & Security Lead	Member 3 - Rawan	Week 5	Week 6							
		Implement Pinia/Vuex & Chart.js	Frontend Lead	Member 1 - ZengLiting	Week 5	Week 6							
		Full System Integration Testing	DevOps, Mobile & Integration Lead	Member 4 - Zineb	Week 5	Week 6							
5	Complex Features, Mobile Build & Final Polish	Develop Streaks, Badges & Challenge Logic	Backend & API Lead	Member 2 - Mohammed	Week 6	Week 7							
		Capacitor Android Application Wrap	DevOps, Mobile & Integration Lead	Member 4 - Zineb	Week 6	Week 7							
		Cloud Deployment & Public URL setup	DevOps, Mobile & Integration Lead	Member 4 - Zineb	Week 7	Week 7							
		Finalize Documentation & PR3/PR4 Demo Prep	All Members	Group 2	Week 7	Week 7							

Figure 11: Project Development Plan GanttChart

Timeline Summary:

- **Weeks 1–2:** UI/UX Design, MySQL Database mapping, static HTML/CSS build, and PHP Slim environment setup.
- **Weeks 3–4:** Scaffold Vue 3 SPA, connect mock JSON APIs via Fetch/Axios, and develop server-side CRUD RESTful endpoints.
- **Weeks 5–6:** Implement carbon calculation logic, integrate JWT authentication, and connect the Vue frontend to the live PHP API.
- **Week 7:** Cloud deployment, Capacitor Android wrapping, and final integration testing for the live presentation.

2. Milestone Targets

✓ **Week 11:**
PR1 Submission

Project Proposal (PR1): Problem statement, objectives, scope, and target users for GreenStep. Includes UI mock-ups, MySQL ER diagram, technology stack definition, this 7-week milestone plan, and the team role split.

✓ **Week 13:**
PR2 Interim Build Submission

Project Interim Progress (PR2): Static responsive UI complete and Vue refactoring underway. Mock JSON data layers connected via Fetch/Axios. In addition to a demonstration of an **End-to-End Feature such as** the Daily Log feature; logging a daily transport activity and seeing the mock carbon footprint update.

✓ **Week 15:**
PR3 Final Integrated Product & PR4 Live Presentation Submission

Project Final Product (PR3): A fully integrated, working product using Vue 3, PHP Slim 4, MySQL (with PDO), and JWT. The application will be deployed at a public URL, with the Android build running on an emulator or physical device along with documentation.

Project Demo (PR4): A 15-minute live demo with a Q&A where every team member will speak. We will demonstrate the working app and deliver our entrepreneurial pitch (e.g., introducing a B2B tier for corporate sustainability programs).

VIII. Role Split

1. Project Overall Role Split

In this project, the roles were split among the team members in the following manner:

Member Name	Role	Primary Duties
Zeng Liting (A24CS4073)	Member 1 — Frontend Lead	Vue project scaffolding, design system, components, routing, state management, mobile-first CSS, accessibility, screen-by screen UI build.
Mohammed Mohsen Alsakkaf (A23CS4026)	Member 2 — Backend & API Lead	PHP Slim project setup, RESTful endpoint design, business logic, JSON request/response shaping, error handling, API documentation.
Rawan MohamedSalih Magzob (A24CS4066)	Member 3 — Database & Security Lead	ER diagram, MySQL schema, PDO data-access layer, prepared statements, JWT issuance/verification, input validation, XSS escaping, session/role policy.
Ennabili Zineb (A24CS4057)	Member 4 — DevOps, Mobile & Integration Lead	Cloud deployment of API + DB + frontend, CI/CD basics, Capacitor Android wrap, environment config, integration testing, demo orchestration.

2. Project Proposal Role Split

Especially for this proposal, the sections were assigned among team members in the following manner:

Member Name	Role	Assigned Section(s)
Zeng Liting (A24CS4073)	Member 1 — Frontend Lead	III (Section 1), V
Mohammed Mohsen Alsakkaf (A23CS4026)	Member 2 — Backend & API Lead	II, III (Section 2)
Rawan MohamedSalih Magzob (A24CS4066)	Member 3 — Database & Security Lead	I, III (Section 3), IV
Ennabili Zineb (A24CS4057)	Member 4 — DevOps, Mobile & Integration Lead	III (Section 4), VI, VII, VIII